KUM DONALSIEN AKWO

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SUMMARY

Computer Engineering graduate specializing in networks and security, with two years of hands-on network engineering and security operations at Telnet Cameroon and a record of building and shipping real security tools. Strong in intrusion detection, network security and applied machine learning for defense. Program Delegate for the Computer Engineering department.

EDUCATION

University of Buea, Faculty of Engineering and Technology Sept 2026 to present

B.Eng in Computer Engineering, one year programme. Currently enrolled.

College of Technology, University of Buea Sept 2023 to Jul 2026

B.Tech Computer Engineering (First Class Honors), option in Computer Networks and Systems Maintenance. CGPA 3.73/4.0. Coursework completed; degree conferral December 2026.

Relevant coursework: Data Structure and Algorithm, Algorithm and Complexity, Operating Systems, Computer Architecture, Computer Networks I and II, Data Security and Integrity, Information Systems Security, Introduction to Data Analysis and Machine Learning.

GCE Advanced Level, Cameroon June 2023

Biology, Chemistry, Pure Mathematics with Mechanics, Physics, Computer Science.

RESEARCH AND PROJECTS

Bastion IDS, Hybrid Intrusion Detection System (2025 to 2026, capstone).

- Four-layer detection: a signature engine (ET-Open ruleset, 47,357 signatures), a machine learning ensemble (Random Forest, XGBoost, CatBoost), a deep neural network specialist (residual MLP) and an anomaly detection engine, built as a standalone desktop application for real-time network monitoring.
- Stack: Python, FastAPI, React, Scikit-learn, XGBoost, CatBoost, Keras/TensorFlow, Snort, Wireshark.

Stop Arnaque 237 and CheckAm (2026).

- Released Stop Arnaque 237, an open labeled dataset of real scams in Cameroon (public, with a DOI). Built CheckAm on top of it, a live service where ordinary people check whether a message is a scam and fraud analysts review the wider campaigns behind those messages.
- **Stillwatch, inactivity detection for people living alone (2026).**
- Tells somebody who left the house from somebody who stopped moving in it, by classifying each camera as a way out or an interior room and reading the sequence rather than a threshold. Learns each household's own rhythm hour by hour. Live at stillwatch.tech under Apache 2.0, with 12 test suites and 5 static checks. Built on the Ring Partner API with Amazon Bedrock, SNS, CloudWatch, Lambda and EventBridge. Stack: Python, Flask, PostgreSQL, AWS.
- **ring-event-simulator, testing tool for the Ring API (2026).**
- A standalone generator of realistic multi-week household event streams, so inactivity detection can be built and tested without hardware or API access. No dependencies, deterministic from a seed, with its own test suite and CI. Published separately under MIT.
- **Countersign, a guard for AI agents moving money (2026).**
- Checks what an autonomous agent is about to do against the merchant's own rules before the payment leaves and explains any refusal in plain words. Stack: Python, deployed on Render.

FairGlass, Proof of Fair AI Hiring (2026, MLH hackathon).

- Built on the Midnight blockchain. A Compact smart contract proves an AI hiring decision followed an agreed fairness policy and issues a verifiable receipt without exposing candidate data. Contributed the backend, scoring logic and contract bridge.

Evaluation of Network Monitoring Tools (2024).

- Evaluated IDS/IPS and network monitoring tools and produced a formal analysis and recommendations report. Wireshark, Snort, Suricata, Nmap, Security Onion, Kitsune.

PUBLICATION

Tatsachou Ngningaye D., Kum Donalsien A., Gamom Ngounou Ewo R. C., Boum Alexandre T. “Bastion IDS: Layer Contribution, Threshold Sensitivity, and Consensus Rule Design in a Deployed Hybrid Intrusion Detection System.” Journal of Cyber Security Technology (Taylor & Francis), 2026. Submitted.

EXPERIENCE

Telnet Cameroon Aug 2024 to Present

Senior Intern and Intern Instructor (Jul to Sep 2025); Support Staff (Oct 2024 to Present); Intern (Aug to Sep 2024).

- Configured enterprise networks: Cisco switches and routers, Fortigate firewalls, routing protocols (OSPF, BGP, IS-IS, RIP), wireless access points and secure remote access.
- Deployed IP surveillance systems and Starlink satellite internet infrastructure on client sites.
- Led training and lab sessions for a new batch of junior interns in network configuration, troubleshooting and cybersecurity.
- Recognized as Best Intern for the Aug to Sep 2024 batch by Telnet Cameroon management.

TECHNICAL SKILLS

Cybersecurity: Kali Linux, Wireshark, Nmap, Snort, Suricata, IDS/IPS, ethical hacking, penetration testing, Fortigate, security operations, threat intelligence, MITRE ATT&CK.

Networking: Cisco IOS, routers and switches, OSPF, BGP, IS-IS, RIP, network design, wireless, IP addressing.

Cloud (AWS): EC2, S3, VPC, IAM, CloudFront, CloudWatch, Route 53, cloud security.

Programming: Python, JavaScript, C, MySQL.

AI and Machine Learning: Scikit-learn, Keras/TensorFlow, Random Forest, XGBoost, CatBoost, Isolation Forest, Autoencoders, deep neural networks.

Tools: FastAPI, Flask, React, Docker, Git, VirtualBox.

CERTIFICATIONS

- Certified in Cybersecurity (CC), ISC2 (2026 to 2029).
- Fortinet Certified Fundamentals (FCF) in Cybersecurity, Fortinet (2026 to 2028).
- Cisco Introduction to Cybersecurity (2025); Cisco Networking Academy Capture the Flag 2026.

LEADERSHIP AND ACTIVITIES

- Program Delegate, Computer Engineering department (option: Computer Networks and Systems Maintenance), liaison between students and university administration.
- 237HackFest Capture the Flag (Hack The Box Cameroon); MLH x Midnight 48-hour hackathon.

AWARDS

- Best Intern, Telnet Cameroon (2024). Leadership award, College of Technology. Academic excellence recognition.

LANGUAGES

English (fluent), French (basic).